THE CANONICAL MORPHISM INTO THE RELATIVE SPECTRUM OF THE PUSHFORWARD

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Allowed standard prerequisites (editorial).

- Represented functor. definition-relative-spec with representability lemma-spec (carded this batch). The relative spectrum represents the functor F, so the counit pair over X produces the morphism can.
- Affine description. lemma-spec-affine (same chapter, judged non-self-contained this batch, not carded). Identifies the relative spectrum over affine opens with the spectrum of the sections.
- Morphism into an affine scheme. schemes-lemma-morphism-into-affine (cross-chapter, named). The affine-restriction morphism f^{-1}(U) to Spec of the sections.

Source reference. Original lemma 01LT: sources/stacks-constructions/pinned-original-source.tex, source lines 709–709. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01I1: focused-seeds/geometry/S9-cons-canonical-morphism/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01LC: focused-seeds/geometry/S9-cons-canonical-morphism/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a quasi-compact and quasi-separated morphism of schemes. By Schemes, Lemma 01LC the sheaf $f_*\mathcal{O}_X$ is a quasi-coherent sheaf of $\mathcal{O}_S$-algebras. There is a canonical morphism*

$$can : X \longrightarrow \underline{\mathrm{Spec}}_S(f_*\mathcal{O}_X)$$

of schemes over S. For any affine open $U \subset S$ the restriction $can|_{f^{-1}(U)}$ is identified with the canonical morphism

$$f^{-1}(U) \longrightarrow \mathrm{Spec}(\Gamma(f^{-1}(U), \mathcal{O}_X))$$

coming from Schemes, Lemma 01I1.

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The morphism comes, via the definition of $\overline{\mathrm{Spec}}$ as the scheme representing the functor F, from the canonical map $\varphi : f^*f_*\mathcal{O}_X \to \mathcal{O}_X$ (which by adjointness of push and pull corresponds to id : $f_*\mathcal{O}_X \to f_*\mathcal{O}_X$). The statement on the restriction to $f^{-1}(U)$ follows from the description of the relative spectrum over affines, see Lemma 01LT. □